MID:1
Subject : DM

Q1. What is Data Mining?(2M)

Data Mining means finding useful information from a large amount of data.
It helps us understand patterns and make better decisions.

Example:
Banks use data mining to find fraud transactions.

OR

Q2. What are Patterns, Trends, and Prediction?(2M)

Pattern:
A pattern is a repeated or regular relationship found in data. It shows how certain items or events are connected.
Example: Customers who buy tea often also buy sugar.

Trend:
A trend is the general direction in which data moves or changes over a period of time. It can show an increase, decrease, or stability.
Example: Online shopping has been increasing every year.

Prediction:
Prediction means estimating future outcomes based on past and present data. It helps in planning and decision-making.
Example: Using last year's sales data to predict next month's sales.

Q3. Explain KDD Process with Diagram.(4M)

KDD (Knowledge Discovery in Databases) is the process of finding useful knowledge from large data.

Steps of KDD:

Data Selection:
Relevant data is selected from large databases.
Only useful data for analysis is chosen.

Data Cleaning:
Errors, duplicate records, and noise are removed.

Missing values are corrected or filled.

Data Integration:
Data from multiple sources is combined.
Different datasets are merged into one consistent dataset.

Data Transformation:
Data is converted into a proper format for mining.
Techniques like normalization and aggregation are applied.

Data Mining:
Algorithms are applied to discover patterns.
Hidden relationships and trends are identified.

Evaluation:
Extracted patterns are checked for usefulness.
Only meaningful results are selected.

Knowledge Presentation:
Final knowledge is presented using reports, graphs, or charts.
Results are shown in an understandable form for decision-making.

Diagram:

Data Sources
↓
Selection
↓
Cleaning
↓
Integration
↓
Transformation
↓
Data Mining
↓
Evaluation
↓
Knowledge

OR

Q4. Describe Importance of Data Mining in Decision Making.(4M)

Data Mining plays an important role in decision making because it helps organizations analyze large amounts of data and find useful information.

Importance of Data Mining:

Better Decision Making:
Data mining provides accurate and meaningful information from large data.
It helps managers make smart and data-based decisions.

Understanding Customer Behavior:
It helps businesses understand customer needs, buying habits, and preferences.
Companies can improve products and services based on this analysis.

Fraud Detection:
Data mining identifies unusual patterns or activities.
It is widely used in banks and financial institutions to detect fraud.

Risk Management:
Companies can predict possible risks using past data.
This helps them take preventive actions in advance.

Improves Business Strategy:
It supports better planning for marketing, sales, and growth.
Businesses can create effective strategies for future success.

Q5. Identify real-life example where Data Mining can be applied in Banking.(5M)

Data Mining is widely used in the banking sector to improve security, customer service, and decision-making.

Real-Life Example:
In banks like State Bank of India, data mining is used in the following ways:

Fraud Detection:
It helps identify unusual or suspicious transactions.
If a transaction pattern is different from normal behavior, the system gives an alert.

Loan Approval:
Data mining predicts whether a customer is likely to repay the loan.
It analyzes income, past loan records, and financial behavior.

Credit Risk Analysis:
Banks check the customer's credit history before giving loans.
This reduces the risk of loan default.

Customer Segmentation:
Customers are grouped based on income, spending habits, and preferences.
This helps banks provide personalized services and offers.

Q6. Define Data Warehouse.(2M)

A Data Warehouse is a centralized system that stores large amounts of integrated and historical data collected from different sources.
It is mainly used for analysis, reporting, and decision-making, not for daily transactions.

The data in a data warehouse is:

Subject-oriented
Integrated
Time-variant (stores historical data)
Non-volatile (data is not frequently changed)

Example:
A company stores 5–10 years of sales data in a data warehouse to analyze yearly growth and business performance.

OR

Q7. Define Data, Information, and Knowledge. (2M)

Data, information, and knowledge are related terms used in data processing and decision-making.

1.Data:
Data are raw facts and figures that do not have any meaning by themselves.
They can be numbers, text, symbols, or observations.
Example: 85, 90, 78 (marks of students).

2.Information:
Information is processed or organized data that has meaning.
When data is summarized or calculated, it becomes useful.
Example: Average marks = 84.3.

3.Knowledge:
Knowledge is the understanding gained from information.
It helps in making decisions.
Example: Since the average marks are high, students performed well.

Q8. Explain how data becomes information with example.(3M)

Data is a collection of raw facts and figures. By itself, data does not have meaning. When data is processed, organized, or analyzed, it becomes information.

Process:

Collection of Data:
In this step, raw facts and figures are gathered from different sources.
The data can be collected through surveys, records, websites, or transactions.

Processing:
The collected data is organized and calculations are performed.
It may include sorting, classifying, summarizing, or computing totals.

Interpretation:
The processed results are analyzed to understand their meaning.
This step helps in drawing conclusions and making decisions based on the data.

Example:
Data: 100, 150, 200 (daily sales in rupees).
Processing:Total sales = 100 + 150 + 200 =450
Information: Total sales for three days is ₹450.

Now it becomes meaningful and useful for decision-making.

OR

Q9. Difference Between Database and Data Warehouse.(3M)

1.Purpose
Database: Used for daily transactions.
Data Warehouse: Used for analysis and decision-making.

2.Type of Data
Database: Stores current and operational data.
Data Warehouse: Stores historical data.

3.System Type
Database: Supports OLTP.
Data Warehouse: Supports OLAP.

4.Data Update
Database: Frequently updated (insert, update, delete).

Data Warehouse: Rarely updated.

5.Users
Database: Used by clerks and operational staff.
Data Warehouse: Used by managers and analysts.

6.Query Type
Database: Simple and fast queries.
Data Warehouse: Complex analytical queries.

7.Data Source
Database: Usually single application.
Data Warehouse: Data collected from multiple sources.

8.Size of Data
Database: Smaller in size.
Data Warehouse: Very large in size.

Q10. Classify OLTP and OLAP.(4M)

OLTP and OLAP are two different types of data processing systems used in organizations.

1.OLTP (Online Transaction Processing)
Used for managing daily business transactions.
Handles a large number of simple queries.
Data is frequently updated (insert, update, delete).
Focuses on speed and accuracy.
Stores current and operational data.

Examples:
ATM transactions, railway ticket booking, online shopping systems.

2.OLAP (Online Analytical Processing)
Used for data analysis and decision-making.
Handles complex queries and large datasets.
Data is not frequently changed.
Stores historical data.
Used by managers and analysts.

Examples:
Sales analysis reports, yearly profit analysis, business forecasting.

Q11. Define Decision Tree and Naïve Bayes.(3M)

1.Decision Tree

A Decision Tree is a supervised learning algorithm used for classification and prediction. It represents decisions in the form of a tree structure.
It starts with a root node (main attribute).
Data is split into branches based on conditions.
Each branch leads to another decision node or a leaf node.
The leaf node gives the final result (class label).

Example:
It can be used to predict whether a person will buy a product based on age and income.

2.Naïve Bayes

Naïve Bayes is a classification algorithm based on Bayes Theorem and probability.
It assumes that all features are independent.
It calculates the probability of each class.
The class with the highest probability is selected.

Example:
Used in spam email detection and text classification.

Q12. Explain Working of Decision Tree.(3M)

A Decision Tree works by splitting data into smaller parts based on certain conditions to make a decision.

Working Steps:

1.Select Best Attribute
The algorithm selects the best attribute using measures like entropy or Gini index.

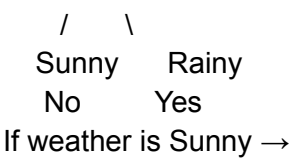
2.Create Root Node
The selected attribute becomes the root node of the tree.

3.Split the Data
Data is divided into branches based on the values of the selected attribute.

4.Repeat the Process
For each branch, again select the best attribute and split the data.

5.Create Leaf Node
When no further splitting is possible, a leaf node is created which gives the final decision.

Example:
Suppose we want to decide whether to play cricket based on weather.



If weather is Sunny → Do not play
If weather is Rainy → Play

Q13. Apply K-Means Clustering on Small Dataset.(4M)